

Demo: Homopolar Motor

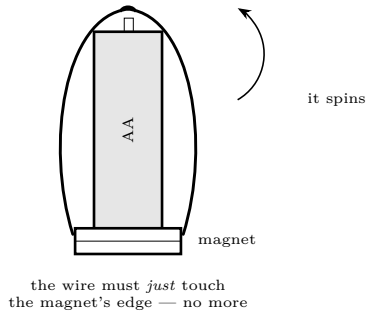
An electric motor with three parts and no switch

*Five minutes, and it will run until the
battery dies*

Lesson Demo A · 5 minutes · after Lesson 4 — battery only, completely safe

What you need

- One AA battery
- One neodymium disc magnet, 10–12 mm
- About 20 cm of bare copper wire, 18–20 AWG
- Pliers



Predict first

There is no switch, no commutator and no coil — three parts and one loop of wire. Before you build it: what is actually pushing on the wire, and which direction will it turn? Then, once it spins: what happens if you flip the magnet over?

Why it turns Current flows down the wire. The wire sits in the magnet's field. A current in a magnetic field feels a sideways force — and because the field is vertical and the current has a horizontal part, the force pushes it around in a circle.

The rule Point the fingers of your right hand along the current and curl them into the field: your thumb gives the force. Reverse either the current or the magnet and it reverses.

Why so simple Every other motor needs a commutator to keep flipping the current. This one never needs to, because the geometry means the force is always in the same rotational direction.

Build it

1. Stick the **magnet to the flat negative end** of the battery. Stand the assembly on the magnet.
2. Bend the wire into a shape that **rests in a dimple on the positive terminal** and hangs down both sides, with the two ends just brushing the outer edge of the magnet. Symmetry matters more than neatness.
3. Let go. It should spin immediately, often at several hundred revolutions per minute.
4. If it does not: the ends are pressing too hard (friction), not touching at all (no circuit), or the wire is not balanced. Adjust one thing at a time.

Break it

Flip the magnet

Reverses the field, so it reverses the spin. Predict before you do it.

Flip the battery

Reverses the current — same result. Now: what if you flip *both*?

Two magnets

Stronger field, more force, faster spin. Also a hotter wire.

Shape it

A heart, a spiral, a cage. Once it works, the shape is a design problem and the boys will spend an hour on it.

Feel the wire

It gets warm. This is a dead short across a battery with almost no resistance — most of the energy becomes heat, not rotation. Do not leave it running for long, and do not use a lithium cell.

Why this one first

This is the cheapest “oh, wow” in the whole course and it is the entire principle of every electric motor ever built: current in a magnetic field feels a force. A washing machine, a drill and a Tesla coil’s primary all rest on the sentence this demonstrates in five minutes with three parts.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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